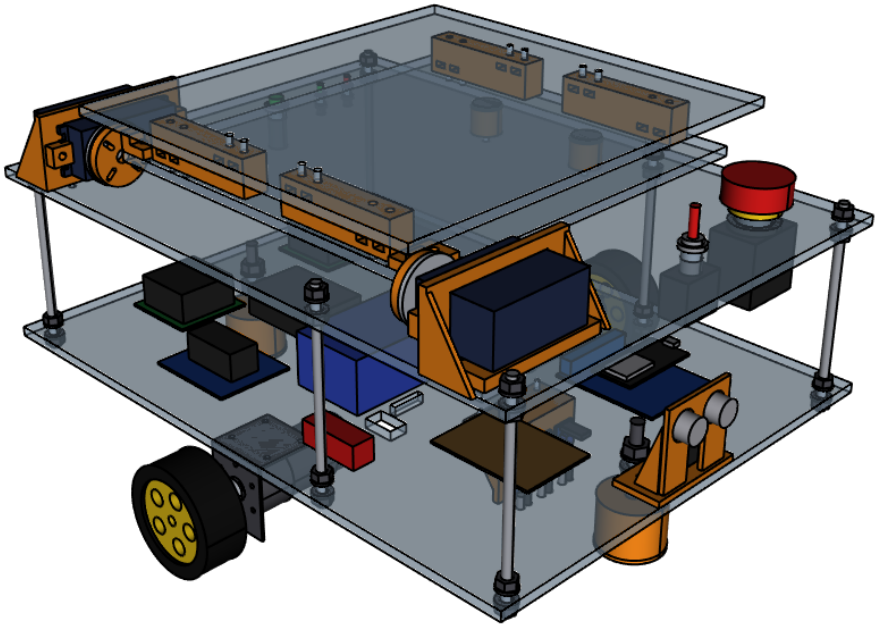
Cargo robot v2

Assembly instructions and list of components



Graduation Project II - 12 November 2026

Contents. 1 Components (bought, laser-cut, 3D-printed). 2 Tools. 3 Assembly, step by step, with a picture of every stage. 4 Final checks. Then drawing sheets 01 (general arrangement), 02 (layout of the electronics bay) and 07 (the tilting bed), at A3. The wiring is in CargoRobot_v2_connections.pdf.

All positions are in millimetres from the centre of the plates: +X is forward, +Y is left. The bed tips to the robot's RIGHT.

1. Components

1.1 Bought parts

No.	Group	Part	Qty	Make / buy
1	Compute	ESP32 DevKitC 38-pin (ESP32-WROOM-32, USB) + 38-pin screw-terminal breakout board	1	buy (get 2)
2	Drive	JGB37-520 12 V 110 RPM gear motor SET: motor with Hall encoder (6 wires) + steel bracket + hex coupling + 65 mm wheel	2	buy
3	Drive	BTS7960 (IBT-2) 43 A motor driver	2	buy
4	Drive	Ball caster: cup (caster_cup.stl) + ring (caster_ring.stl)	2	3D print PETG
5	Drive	Steel bearing ball 25.4 mm (1 inch), or a 25 mm glass marble	2	buy (get 3)
6	Drive	M8 x 50 bolt + 2 M8 nuts + 2 M8 washers (one set per caster)	2	buy
7	Sensors	IR obstacle-avoidance module, 3 pins (VCC, GND, OUT), one blue pot (FC-51 / HW-201 type)	3	buy (get 4)
8	Sensors	Line sensor bar (line_sensor_bar.stl)	1	3D print PETG
9	Sensors	HC-SR04 ultrasonic distance sensor	1	buy (get 2)
10	Sensors	Ultrasonic sensor bracket (sonar_bracket.stl)	1	3D print PETG
11	Tilting bed	Cargo plate 260x280x6 (cargo_plate.dxf)	1	laser-cut 6 mm acrylic
12	Tilting bed	Bed riser (bed_riser.stl)	4	3D print PETG
13	Tilting bed	Tilting plate 236x247x6 (tilt_plate.dxf)	1	laser-cut 6 mm acrylic
14	Tilting bed	Servo DS5160, 60 kg.cm, metal gears, 180 deg, 6-8.4 V, 65 x 30 x 48 mm	2	buy
15	Tilting bed	Aluminium DISC servo horn for the servo's spline (25T on most DS5160), 38 mm or smaller (not the 4-arm horn: its arms hit the deck)	2	buy
16	Tilting bed	Servo mount (servo_mount_front.stl + servo_mount_rear.stl)	2	3D print PETG
17	Tilting bed	Servo ear clamp (servo_clamp.stl)	4	3D print PETG
18	Tilting bed	Horn adapter (horn_adapter_front.stl + horn_adapter_rear.stl)	2	3D print PETG 100 %
19	Tilting bed	Bed rest post (bed_rest.stl) + 18 mm self-adhesive rubber pad 2 mm	2	3D print PETG
20	Tilting bed	Buck converter XL4016 8 A, set to 7.4 V for the servos	1	buy
21	Power	3S LiPo 11.1 V 5000 mAh XT60 + balance charger + LiPo bag + voltage alarm	1	buy
22	Power	XL4015 5 A buck converter (5.1 V for the ESP32 and sensors)	1	buy
23	Power	Inline fuse holder + 10 A fuses, WAGO 221 connectors, XT60 pairs	1	buy
24	Power	Emergency stop 22 mm panel-mount, latching, NC contact block (must clamp a 6 mm panel)	1	buy
25	Power	Toggle switch 12 mm panel-mount, 12 V DC >= 15 A, thread >= 9 mm (6 mm deck)	1	buy
26	Controls	Push button 12 mm panel-mount, momentary, normally open (start), thread >= 9 mm	1	buy
27	Controls	5 mm LED green + red, with panel clips for a 6 mm panel (or hot glue), + 330 ohm resistors	1	buy
28	Controls	Active buzzer module, 3-pin, 3.3-5 V	1	buy
29	Chassis	Bottom plate 400x300x6 (bottom_plate.dxf)	1	laser-cut 6 mm acrylic
30	Chassis	Top deck 400x300x6 (top_deck.dxf)	1	laser-cut 6 mm acrylic
31	Chassis	M6 threaded rod, cut to 130 mm	6	cut from 1 m rod
32	Chassis	M6 nut + M6 washer	24	buy
33	Hardware	M4: x10 (bed rests, 2), x12 (servo mounts, 6, NO washer), x16 (servo ear clamps, 4), x20 (horn adapters 4 + risers from below 8), x20 COUNTERSUNK (risers through the cargo plate, 8); 32 M4 nuts; 32 M4 washers	1	buy
34	Hardware	M3: x8 (servo horns, 8), x8 countersunk (caster rings, 6), x12 + nut (motor brackets 8, holes drilled by hand; IR modules 3), x16 + nut (line sensor bar 2, sonar bracket 2); 30 M3 washers	1	buy
35	Hardware	Resistors 1k x1, 2k x1, 10k x5, 2.2k x1, 100 nF x1, small perfboard; VHB tape, velcro strap, zip ties, a little grease (casters)	1	buy

1.2 Laser-cut plates (files in cad/output/dxf/)

File	Material	Size (mm)	Qty	Hole table
bottom_plate.dxf	6 mm acrylic	400 x 300	1	sheet 03
top_deck.dxf	6 mm acrylic	400 x 300	1	sheet 04
tilt_plate.dxf	6 mm acrylic	236 x 247	1	sheet 05
cargo_plate.dxf	6 mm acrylic	260 x 280	1	sheet 06

1.3 3D-printed parts (files in cad/output/parts/; PETG, 0.2 mm layers, 4 walls)

File	Qty	Orientation	Infill	Supports
servo_mount_front.stl, servo_mount_rear.stl	1 + 1	lying on the ears' face	40 %	none
servo_clamp.stl	4	flat face down	40 %	none
horn_adapter_front.stl, horn_adapter_rear.stl	1 + 1	round wall on the bed	100 %	on build plate only
bed_rest.stl	2	pocket side down	40 %	none
bed_riser.stl	4	on its 80 x 16 face	40 %	none
line_sensor_bar.stl	1	on the flange's top face	40 %	none
sonar_bracket.stl	1	base down	40 %	none
caster_cup.stl	2	upside down (top face on the bed)	40 %	none
caster_ring.stl	2	flat	40 %	none

Every STL file already lies in its printing position: print it as it opens in the slicer. The servo mounts fit the DS5160's outline (65 x 30 mm body, ears 31 mm above its bottom, from the maker's drawing); they clamp the ears, so the ears' notches don't matter. Use a disc horn of 38 mm or less.

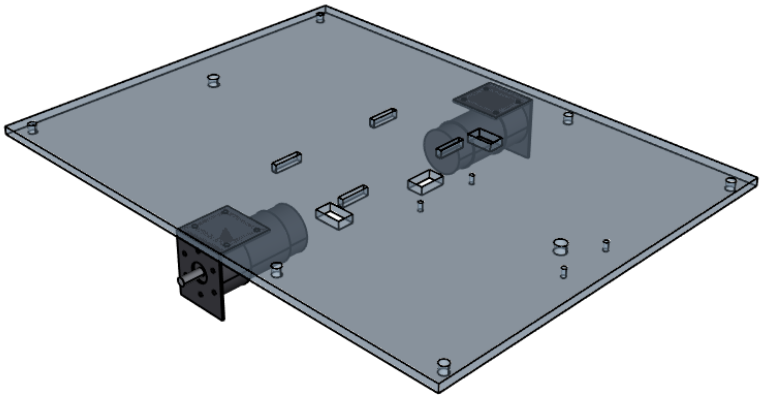
2. Tools

Soldering iron and solder; multimeter with a 10 A range; wire stripper, side cutters, ferrule crimper; hacksaw and file (threaded rod); drill with a 3.5 mm bit (motor brackets: the only holes you drill) and a 90 deg countersink bit (cargo plate); spanners 7, 10 and 13 mm, hex keys, screwdrivers; heat gun; tape measure, square ruler; phone level / angle app; hot glue gun.

3. Assembly, step by step

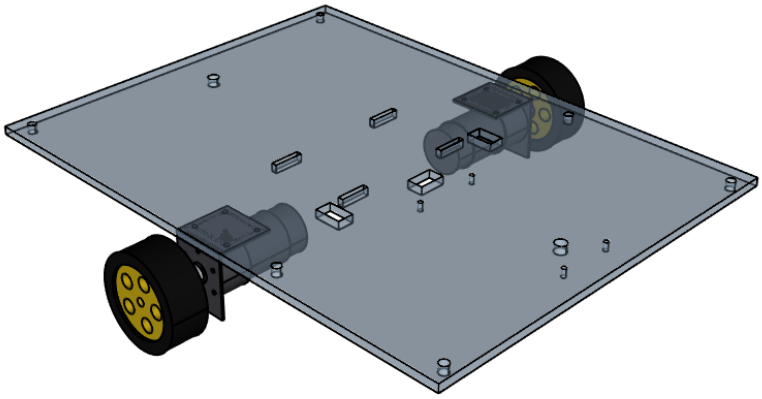
All four plates are 6 mm acrylic: put a washer under every head and nut that touches them, and don't over-tighten - firm with a small spanner is enough. Each picture shows everything fitted so far.

Step 1: Motors under the bottom plate



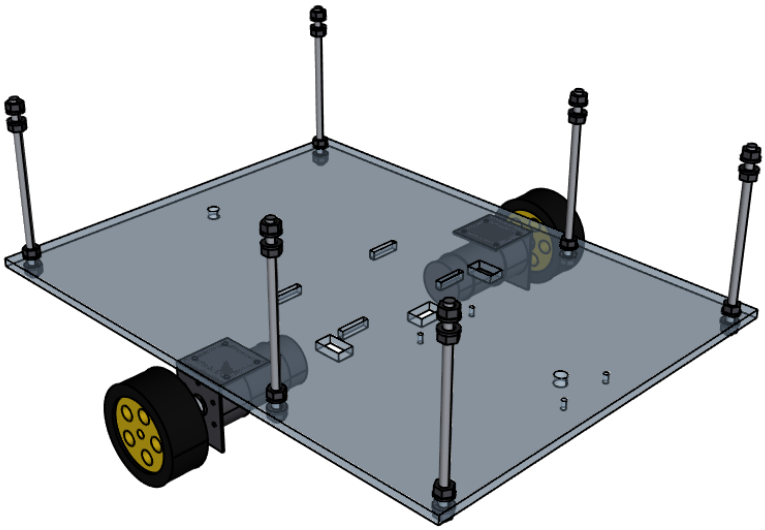
1. Fix each motor in its steel bracket (the set's M3 screws), the shaft through the bracket's upright.
2. Turn the bottom plate upside down. Place each bracket's upright 5 mm in from the side edge, centred at X = 0, the motor pointing inwards. Mark the bracket's base holes and drill them 3.5 mm (acrylic: sharp bit, slowly, a piece of wood underneath). Fix with M3 x 12 + nuts, a washer on the acrylic.
3. Route each motor's 6-wire cable up through the nearest cable slot (X = +40, Y = +-70).

Step 2: Drive wheels



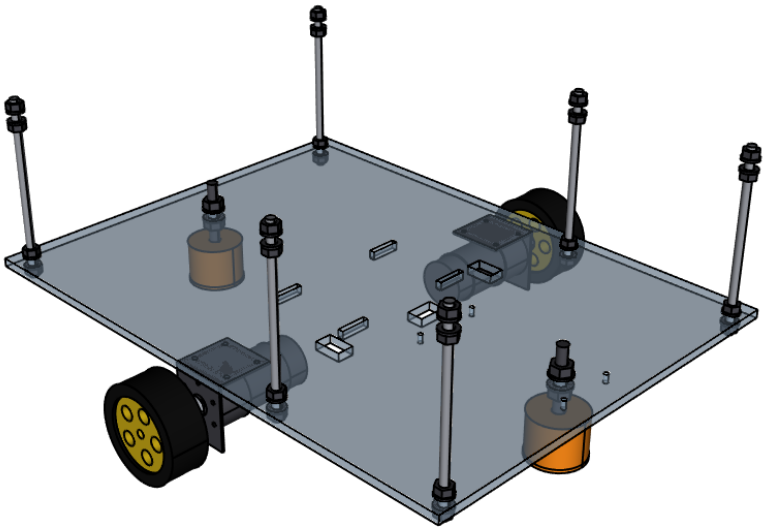
1. Slide the hex coupling onto the motor shaft, its set screw on the flat of the D, a drop of blue thread-locker, tighten.
2. Screw the 65 mm wheel onto the coupling.

Step 3: Threaded rods



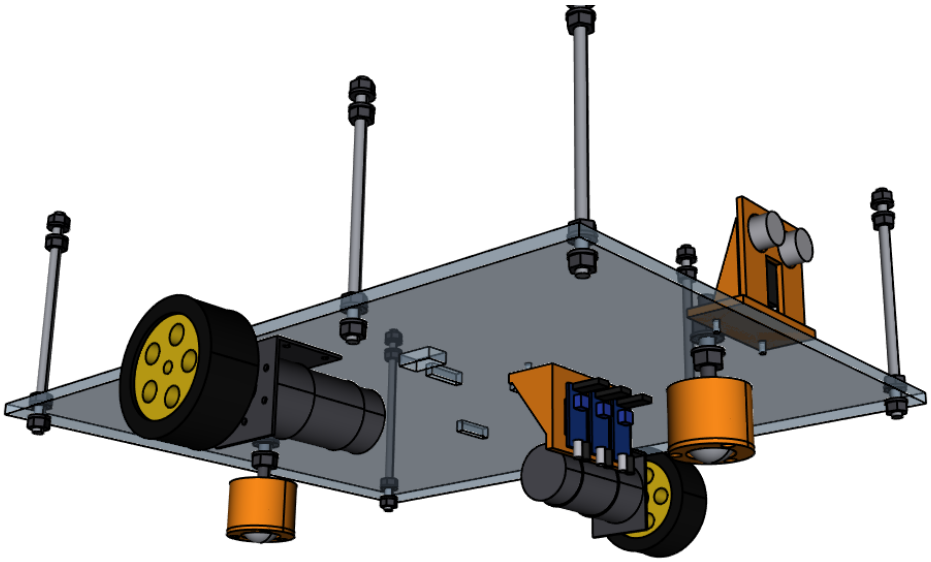
1. Cut 6 pieces of 130 mm from the M6 rod (screw a nut on first, unscrew it over the cut end).
2. Each rod: nut + washer 7 mm from the bottom end, up through the plate, washer + nut on top, tighten.
3. A third nut + washer on each rod with its top exactly 100 mm above the plate (use one gauge for all 6). The deck rests on these washers. Rods at X = -190, +60, +190, Y = +-135.

Step 4: Ball casters



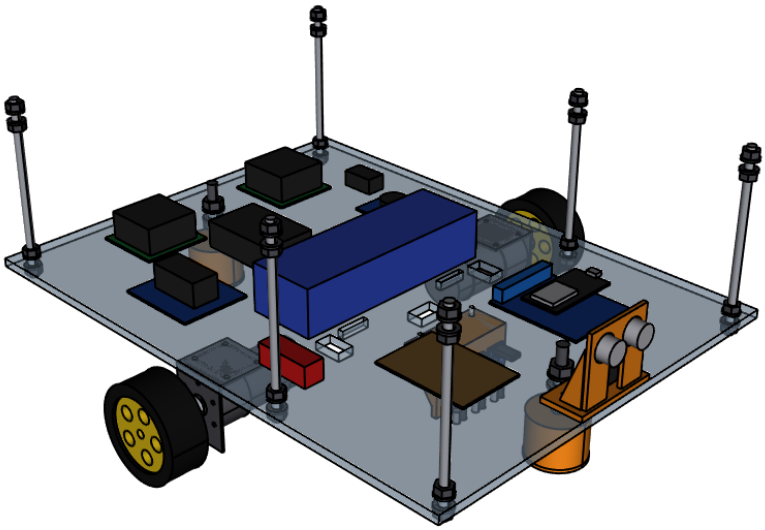
1. Grease the cup's cone. Push an M8 x 50 bolt up through the cup from inside, its head into the hex pocket. Drop the ball in, then screw the ring on (3 x M3 x 8 countersunk).
2. On the bolt: nut + washer, then up through the 8.5 mm hole (X = +-160), washer + nut on top. The height is set later, with all 4 wheels on the floor.

Step 5: Sensors: line sensors underneath, ultrasonic in front



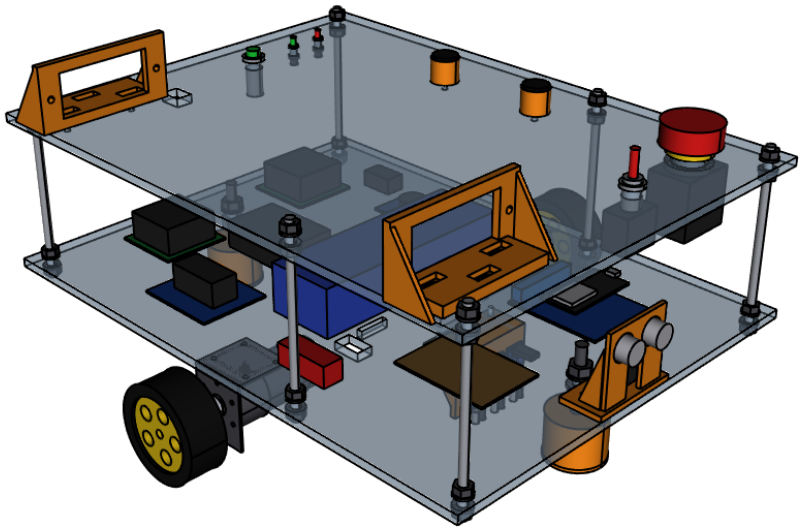
1. If an IR module's 3 pins stick straight out of its end, bend them 90 deg towards its front (pliers).
2. Screw the 3 IR modules to the bar: M3 x 12 through each module's hole and its slot, nut behind; LEDs down, blue screw to the front; module 1 on the robot's LEFT.
3. Bolt the bar under the bottom plate: 2 x M3 x 16 + washer down through the plate (X = 69, Y = +-24), nuts under the flange. Their cables go up through the slot behind the bar.
4. Push the HC-SR04 into the ultrasonic bracket from behind: cans through the windows, board on the 2 ledges, its 4 pins down through the slot. 2 dots of hot glue. Bolt the bracket on top of the plate's front edge (2 x M3 x 16 + nut, washers on the acrylic).

Step 6: Electronics



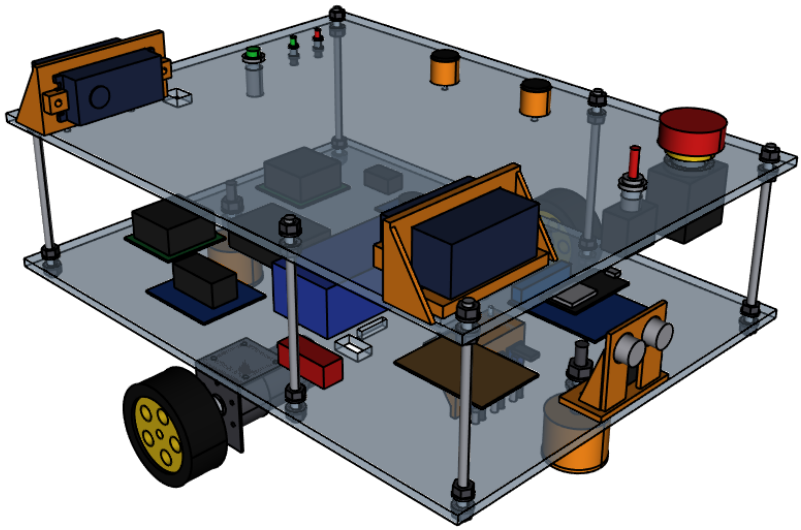
1. Stick every board down with VHB tape, where the picture shows (sheet 02 has the layout with numbers).
2. BTS7960 drivers at the rear, heatsinks up. ESP32 breakout at the front-left, USB socket towards the robot's left side. I/O board (perfboard with the resistors) at the front-right. 5 V buck rear-left, 7.4 V servo buck rear-right, WAGO buses between the battery and the drivers.
3. The battery lies across the middle in 2 velcro straps (through the 4 slots) and slides out sideways.
4. Wiring: weeks 2 and 3 of BUILD_GUIDE.md and CargoRobot_v2_connections.pdf.

Step 7: Deck: controls, servo mounts, bed rests



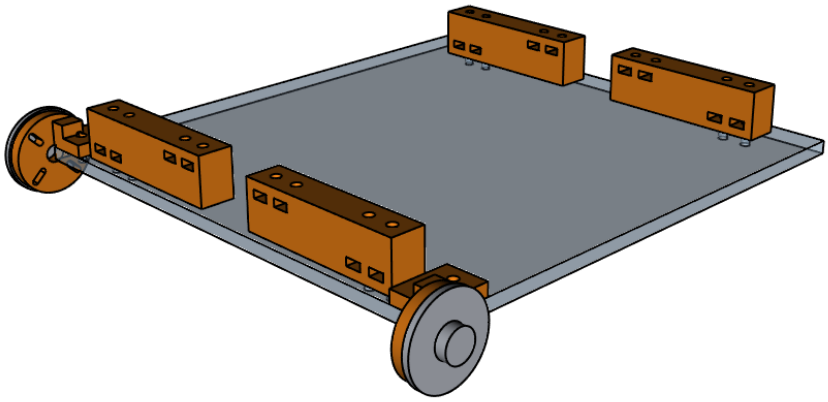
1. Before the deck goes on: E-stop through the 22.5 mm hole (front-left) from above, ring nut below, contact block clipped on, 2 x 40 cm 16 AWG on its NC terminals.
2. Master switch (front, middle) and start button (rear) up through their 12.2 mm holes; LEDs in their clips next to the button (green nearer the button).
3. Servo mounts on the deck's front-right and rear-right corners: an M4 nut in each of the foot's 3 slots, M4 x 12 up through the deck with NO washer (it needs all its length to reach through the nut). Only snug: they are lined up later.
4. Bed rests at X = +-40, Y = +95: an M4 nut in the pocket under each post, M4 x 10 up through the deck; rubber pad on top.
5. Lower the deck onto the 6 rods, washer + nut on top of each, check it is level.

Step 8: Servos



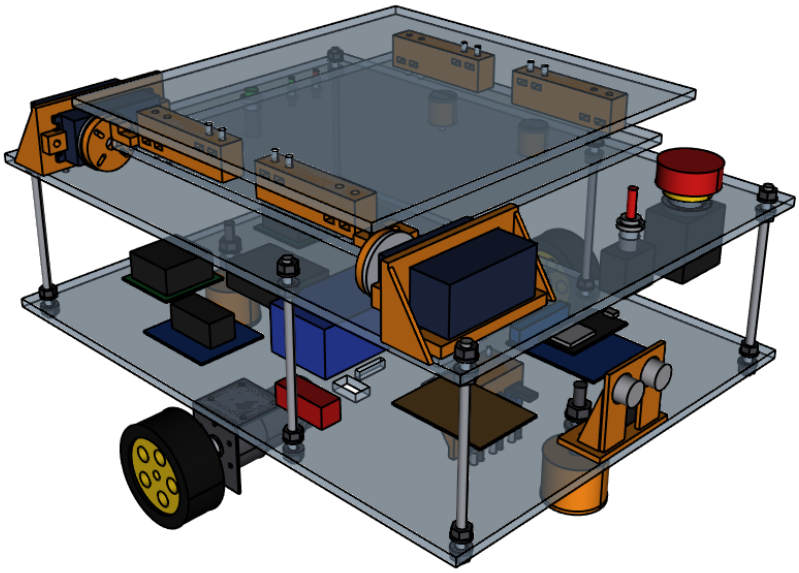
1. Each servo goes into its mount from the bed side, bottom first, until its ears rest on the mount's face; the shaft points to the middle of the robot, on the deck's right edge.
2. 2 ear clamps per servo: M4 x 16 through the clamp and the mount, the clamp's tip on the ear, nut behind.
3. Servo cables down through the cut-outs next to the mounts.

Step 9: The bed (shown on its own)



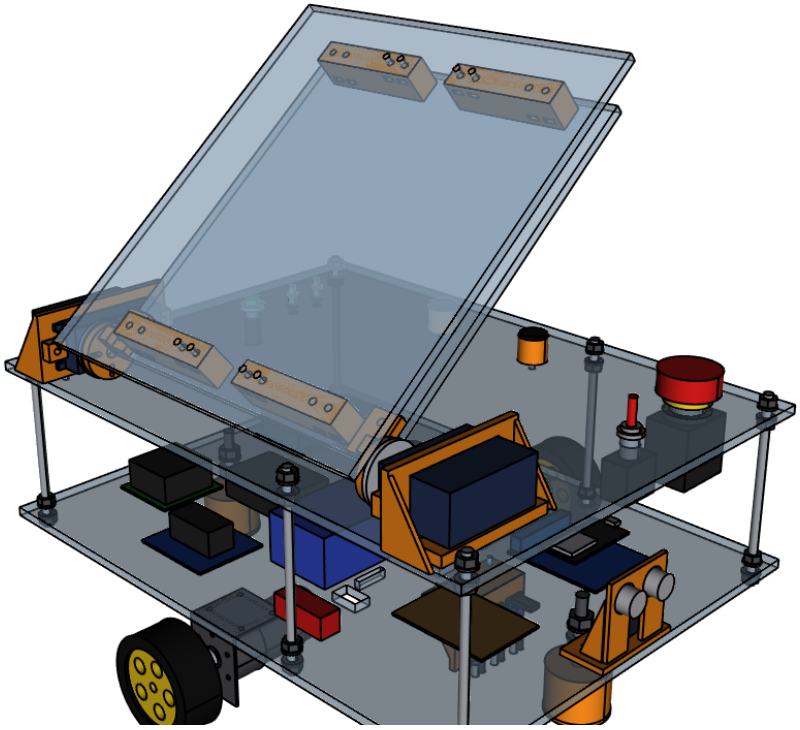
1. Slide an M4 nut into each of the 4 side slots of every riser.
2. Risers on the tilting plate: each riser's outer end over its 2 holes, 2 x M4 x 20 + washer up from below into its nuts. Outer ends towards the front and rear, inner ends towards the middle.
3. Horn adapters on the plate's two right-hand corners (2 x M4 x 20 each, nuts underneath). Then a disc horn on each adapter's round wall: turn it until 4 of its holes sit over the 4 slots, 4 x M3 x 8 (thread-locker).

Step 10: Cargo plate on, bed on the servos



1. Countersink the cargo plate's 8 holes on its top face (90 deg bit, by hand, slowly) so the M4 flat heads sit flush. Lay it on the risers: 8 x M4 x 20 countersunk down into the risers' nuts. No anti-slip mat.
2. Lay the bed on its 2 rest posts and push the two horns onto the servo splines (turn a shaft gently by hand, one tooth at a time, until its horn fits). Fit the horns' centre screws with thread-locker.
3. Line up: slide each servo mount along its slots until its horn sits flat and centred on the adapter, then tighten the 3 bolts under the deck.
4. Set the casters (all 4 wheels on the floor, paper test) and the line sensors (LEDs 5-10 mm above the floor).

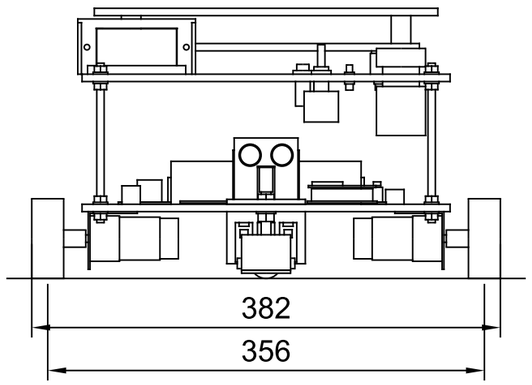
Step 11: Check: the bed swings freely



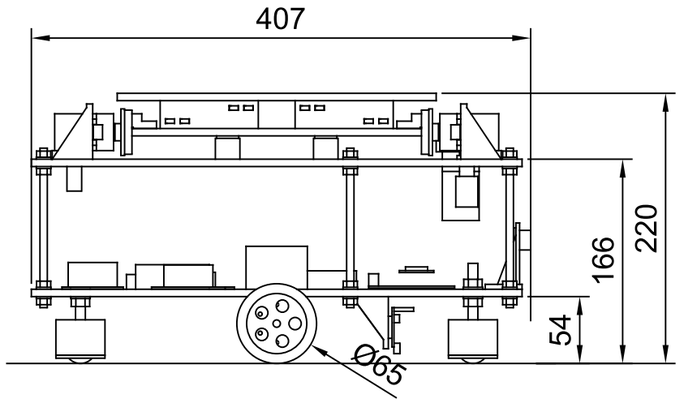
1. Lift the bed's left edge by hand (servos unpowered): it must swing to 45 deg and back without touching anything. The cargo plate's right edge ends just outside the deck.
2. The servo end points are set later with the 'servo' command (BUILD_GUIDE.md, week 4).

4. Final checks

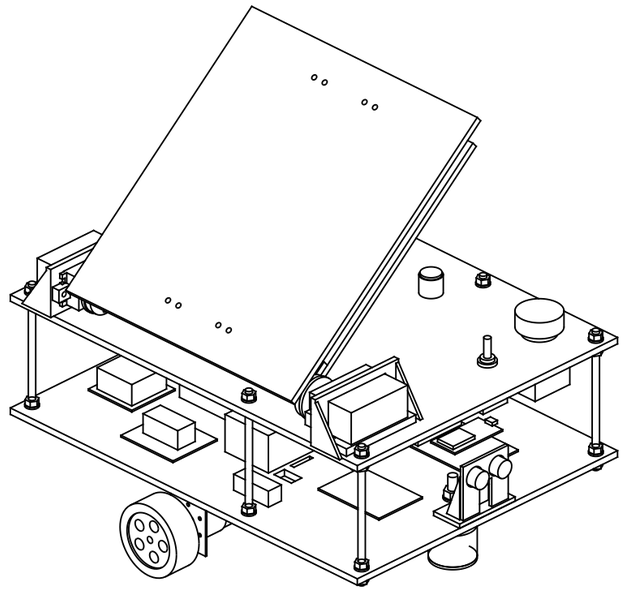
[ ] Push the robot by hand: it rolls straight, turns smoothly, the caster balls roll freely, nothing rubs.
[ ] All 4 wheels on the floor: each caster ball grips a sheet of paper, but a firm tug still pulls it out.
[ ] 3 kg on the cargo plate: nothing bends visibly, no wheel lifts.
[ ] The bed swings from 0 to 45 deg by hand without touching anything, and lies level on its rests.
[ ] Press the E-stop hard: the deck doesn't bend noticeably.
[ ] Take photos of every stage for the report.



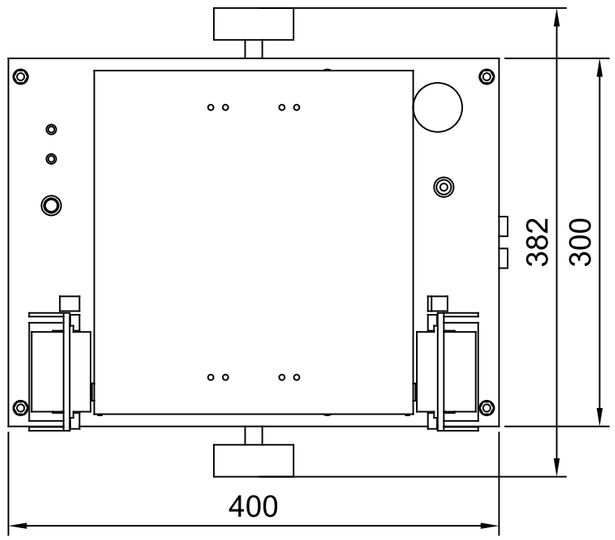
FRONT VIEW



SIDE VIEW (from the right) FRONT -->



ISOMETRIC VIEW - BED TIPPED 40 deg

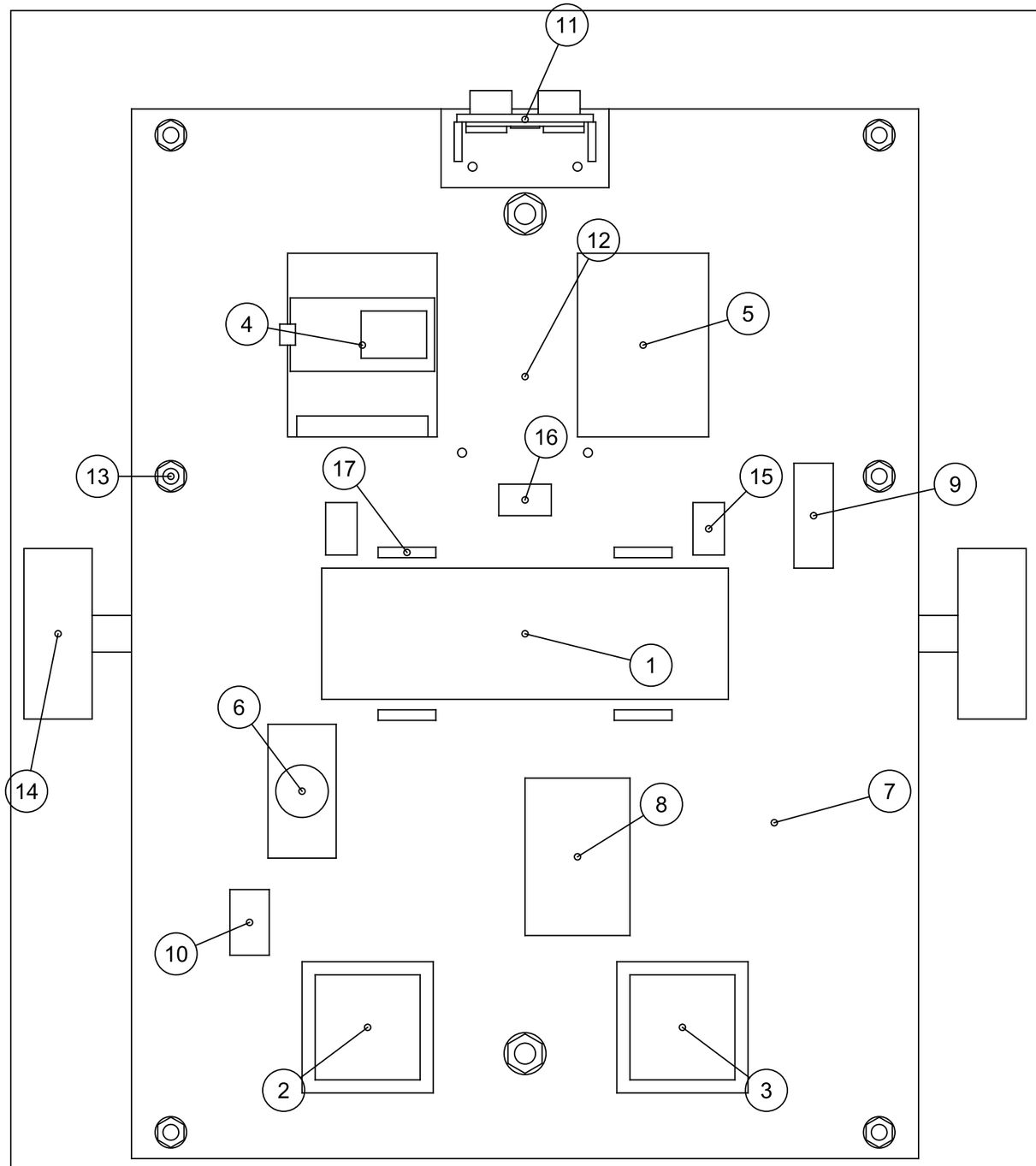


PLAN VIEW FRONT -->

NOTES

1. All dimensions in mm. Weights and loads in kg.
2. Drive: 2 x JGB37-520 12 V 110 RPM motor sets, wheels D65 x 26.
3. Casters: 2 printed ball casters (D25.4 ball) on M8 bolts (height set by nuts).
4. Frame: 6 mm acrylic plates on 6 x M6 threaded rod.
5. Bed: tipped 40 deg to the right by 2 servos on the hinge line.
6. Cargo plate on 4 printed risers; packages up to 3 kg (not weighed).
7. ESP32 (no I/O expander), 3 IR line sensors (under), HC-SR04 (front).
8. Sheets: 01 GA, 02 layout, 03-06 plates, 07 bed, 08-09 prints, 10 BOM.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: GENERAL ARRANGEMENT				DWG No.: CR-V2-001	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:5	SHEET: 1 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:



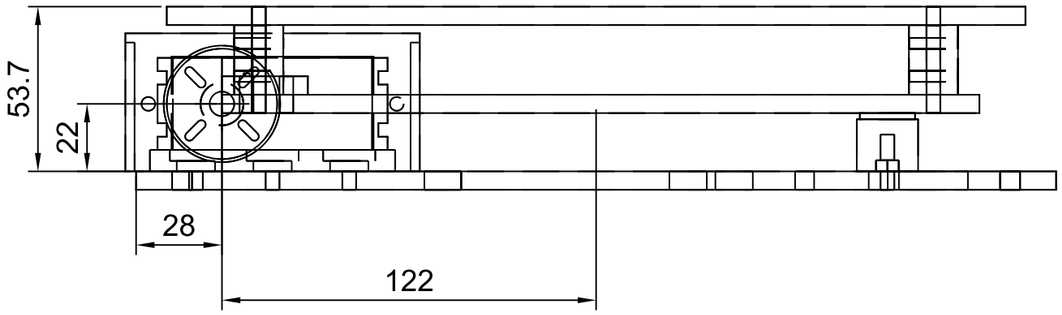
PLAN VIEW - DECK AND BED REMOVED (front of robot at the top)

No.	COMPONENT
1	3S LiPo battery 5000 mAh (lies across, slides out sideways)
2	BTS7960 motor driver - left motor
3	BTS7960 motor driver - right motor
4	ESP32 DevKit on a screw-terminal breakout (USB to the left)
5	I/O board: resistors and dividers (perfboard)
6	Buck converter 12 V -> 5.1 V (ESP32, sensors)
7	Buck converter 12 V -> 7.4 V (bed servos)
8	WAGO power buses (+12 V main / motor / 5 V / GND)
9	Inline fuse 10 A
10	Buzzer
11	HC-SR04 ultrasonic on printed bracket
12	3 IR line sensors on a printed bar (under the plate)
13	M6 threaded rod (x6)
14	Drive wheel D65 (motor under the plate)
15	Motor / encoder cable slot (x2)
16	Line sensor cable slot
17	Battery strap slots (x4)

NOTES

Small boards are held with VHB tape.
Keep 10 cm slack on every wire that goes up to the deck or the bed.
Wiring: see BUILD_GUIDE.md weeks 2 and 3, and the schematics E01-E04.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: LAYOUT - DECK REMOVED				DWG No.: CR-V2-002	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 2 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:



BED FLAT - FROM THE FRONT (front servo not shown)

HOW IT WORKS

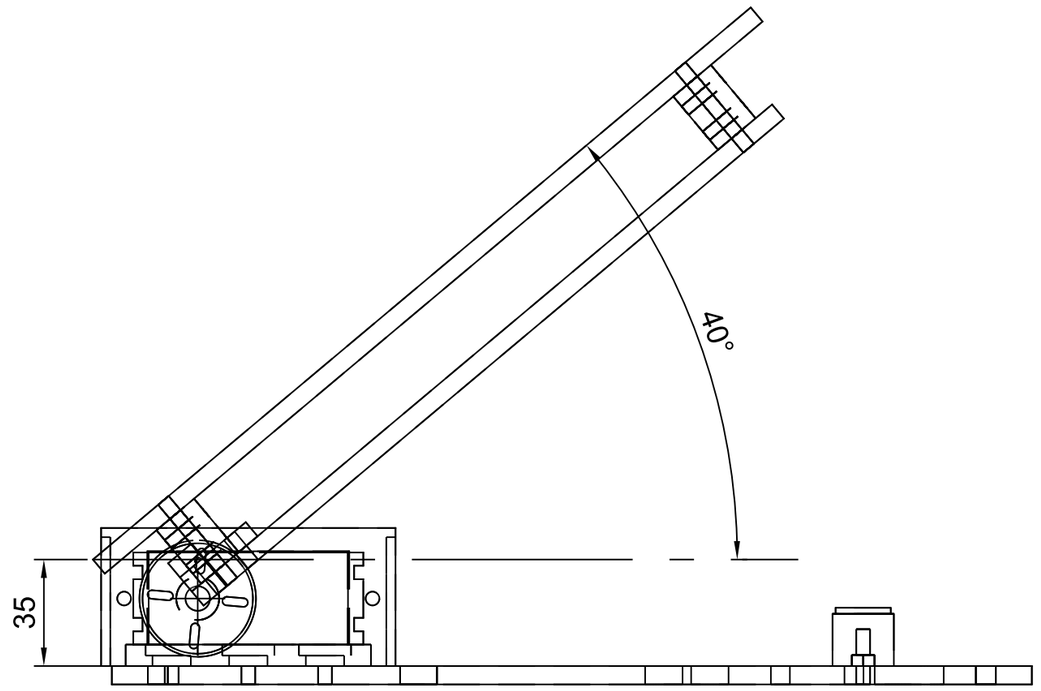
1. Two servos on the deck's right edge share one axis: their shafts ARE the hinge (front and rear end of the bed, mirror images). No hinge, screw or lever.
2. A printed horn adapter on each end of the tilting plate screws to the servo's aluminium disc horn (4 x M3). 4 printed risers carry the cargo plate.
3. Flat, the bed rests on 2 printed posts and the servos are switched off.
4. To unload, the ESP32 turns both servos slowly (20 deg/s) to 40 deg, holds 2 s, then lowers the bed and switches them off.
5. No anti-slip mat: a cardboard box slides on acrylic from about 20 deg.

TORQUE AT THE START OF THE TIP (both servos, 20 cm box)

PACKAGE	TORQUE	OF 2 x 60 kg.cm
1 kg	2.3 N.m	19 %
3 kg	4.7 N.m	40 %
5 kg	7.1 N.m	60 %

SERVOS

Servo DS5160: 60 kg.cm at 7.4 V, 65 x 30 x 48 mm, 75 mm over the ears, ears 31 mm above its bottom (maker's drawing). The ears are clamped, and the mount slides +-4 mm to put the shaft on the axis.
Supply: 7.4 V from its own 8 A buck converter, after the E-stop.
Signal: front servo GPIO 2, rear servo GPIO 12 (schematic E03).
Hinge axis: 188 mm above the floor, 122 mm right of the centre line.



BED TIPPED 40 deg - THE PACKAGE SLIDES OFF TO THE RIGHT

(the right edge of the cargo plate ends up 6 mm outside the deck, 35 mm above it)

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: TILTING BED - SERVO HINGE				DWG No.: CR-V2-007	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 7 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN: